

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

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100/10

Quiz #3

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Closed Notes and Book

Duration: 25 minutes

Fluid at 373 K is to be cooled by a different stream of the same fluid at 293 K. A heat exchanger is available for parallel or counter flow. When the heat exchanger is used with parallel flow, the outlet temperature of the hot stream is 338 K. If the mass flow rates in the hot and cold streams are equal to each other and the heat capacity of the fluid is temperature independent, determine:

- a) The outlet temperature of the cold stream with parallel flow in the heat exchanger, and  
 b) The outlet temperatures of the cold and hot streams when a counter-flow set-up is used.

$$\begin{aligned} T_{h,i} &= 373 \\ T_{c,i} &= 293 \\ T_{h,o} &= 338 \end{aligned}$$

$$a) \quad q = m_h c_p (373K - 338K) = m_c c_p (T_{c,o} - 293)$$

$$\boxed{T_{c,o} = 328K}$$

b)  $q = m c_p \Delta T$   
 From part a, we can find UA:

$$q = UA \Delta T_{\text{em}} = UA 33.66 = m c_p (373 - 338)$$

$$\text{Thus, } UA = 1.04 m c_p$$

Now, for part b we know:

$$m c_p (373 - T_{h,o}) = m c_p (T_{c,o} - 293K)$$

$$100 - T_{h,o} = T_{c,o} - 20$$

$$120 - T_{h,o} = T_{c,o}$$

convert to Celsius

$$100 + 20 - T_{h,o} = T_{c,o}$$

$$120^\circ\text{C} - T_{h,o} = T_{c,o}$$

we also know that

$$m c_p (373 - T_{h,o}) = UA \Delta T_{\text{em}}$$

$$m c_p (373 - T_{h,o}) = 1.04 m c_p \left( \frac{(-20 + T_{h,o}) - (T_{h,o} - 20)}{\ln \left( \frac{-20 + T_{h,o}}{T_{h,o} - 20} \right)} \right)$$

Part a

$$\Delta T_1 = T_{h,i} - T_{c,i} = 80K$$

$$\Delta T_2 = T_{h,o} - T_{c,o} = 10$$

$$\frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2 / \Delta T_1)} = \frac{11}{\ln(10/80)}$$

$$\Delta T_{\text{em}} = ?$$

$$\Delta T_2 = T_{h,i} - T_{c,o} = 100 - T_{c,o} = 100 - 120 = -20$$

$$\Delta T_1 = T_{h,o} - T_{c,i} = T_{h,o} - 20 = T_{h,o} - 20$$

$$X = \frac{\Delta T_2}{\Delta T_1}$$

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1)

$$mcp(100 - T_{h,o}) = 1.04 mcp \left( \frac{\Delta T_1 (X-1)}{\ln(X)} \right)$$

$$\text{as } \lim_{x \rightarrow 1} \left( \frac{\Delta T_1 (x-1)}{\ln(x)} \right) = \frac{\Delta T_1}{1/x} = \frac{\Delta T_1}{1} = \Delta T_1$$

Thus,

$$100 - T_{h,o} = 1.04(\Delta T_1) = 1.04(T_{h,o} - 20)$$

$$100 - T_{h,o} = 1.04 T_{h,o} - 20(1.04)$$

$$120.8 = 2.04 T_{h,o}$$

$$T_{h,o} = 59.2^\circ\text{C}$$

$$T_{h,i}, T_{c,o} = 120 - 59.2 = 60.8^\circ\text{C}$$

$$\text{Therefore } \boxed{\begin{aligned} T_{h,o} &= 273 + 59.2 = 332.2 \text{ K} \\ T_{c,o} &= 60.8 + 273 = 333.8 \text{ K} \end{aligned}}$$

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